UNDERLYING SETS OF LIMITS OF DECENT ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1101-1153, following the statement at 1092-1099.

Editorial note. No load-bearing omissions; the closing parenthesis on quasi-separated / locally separated spaces being decent is transcribed.

Allowed standard prerequisites (editorial).

- Presentation of the limit space. lemma-directed-inverse-system-has-limit (same chapter, cited at ~1104-1105 and ~1114-1115). X = U/R with U = lim U_i and R = lim R_i; also f_0^{-1}W_0 = lim f_{i0}^{-1}W_0.
- Topology of limits of schemes. Limits, Lemma limits-lemma-topology-limit (cross-chapter, cited at ~1117-1118). |U| = lim |U_i| and |R| = lim |R_i| as sets.
- Nonemptiness of directed limits of finite nonempty sets. Categories, Lemma categories-lemma-nonempty-limit (cross-chapter, cited at ~1125-1126 and ~1140-1141). lim S_i and lim T_i are nonempty.
- Decent spaces have finite fibres under etale covers. Decent Spaces, Definition decent-spaces-definition-very-reasonable (cross-chapter, cited at ~1122-1124 and ~1136-1138). S_i and T_i are finite nonempty sets.

Source reference. Original reference 086J: focused-seeds/geometry/S9-spaces-limits-inverse-limit-sets/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03I8: focused-seeds/geometry/S9-spaces-limits-inverse-limit-sets/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 088J: focused-seeds/geometry/S9-spaces-limits-inverse-limit-sets/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03I7: focused-seeds/geometry/S9-spaces-limits-inverse-limit-sets/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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Source reference. Original lemma 07SF: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 878–878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 086V: focused-seeds/geometry/S9-spaces-limits-inverse-limit-sets/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03IJ: focused-seeds/geometry/S9-spaces-limits-inverse-limit-sets/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $X = \lim_{i \in I} X_i$ be the limit of a directed inverse system of algebraic spaces over S with affine transition morphisms (Lemma 07SF). If each X_i is decent (for example quasi-separated or locally separated) then $|X| = \lim_i |X_i|$ as sets.*

Proof. There is a canonical map $|X| \to \lim |X_i|$. Choose $0 \in I$. If $W_0 \subset X_0$ is an open subspace, then we have $f_0^{-1}W_0 = \lim_{i \geq 0} f_{i0}^{-1}W_0$, see Lemma 07SF. Hence, if we can prove the lemma for inverse systems where X_0 is quasi-compact, then the lemma follows in general. Thus we may and do assume X_0 is quasi-compact.

Choose an affine scheme U_0 and a surjective étale morphism $U_0 \to X_0$. Set $U_i = X_i \times_{X_0} U_0$ and $U = X \times_{X_0} U_0$. Set $R_i = U_i \times_{X_i} U_i$ and $R = U \times_X U$. Recall that $U = \lim U_i$ and $R = \lim R_i$, see proof of Lemma 07SF. Recall that $|X| = |U|/|R|$ and $|X_i| = |U_i|/|R_i|$. By Limits, Lemma 086V we have $|U| = \lim |U_i|$ and $|R| = \lim |R_i|$.

Surjectivity of $|X| \to \lim |X_i|$. Let $(x_i) \in \lim |X_i|$. Denote $S_i \subset |U_i|$ the inverse image of x_i. This is a finite nonempty set by the definition of decent spaces (Decent Spaces, Definition 03I8). Hence $\lim S_i$ is nonempty, see Categories, Lemma 086J. Let $(u_i) \in \lim S_i \subset \lim |U_i|$. By the above this determines a point $u \in |U|$ which maps to an $x \in |X|$ mapping to the given element (x_i) of $\lim |X_i|$.

Injectivity of $|X| \to \lim |X_i|$. Suppose that $x, x' \in |X|$ map to the same point of $\lim |X_i|$. Choose lifts $u, u' \in |U|$ and denote $u_i, u_i' \in |U_i|$ the images. For each i let $T_i \subset |R_i|$ be the set of points mapping to $(u_i, u_i') \in |U_i| \times |U_i|$. This is a finite set by the definition of decent spaces (Decent Spaces, Definition 03I8). Moreover T_i is nonempty as we've assumed that x and x' map to the same point of X_i. Hence $\lim T_i$ is nonempty, see Categories, Lemma 086J. As before let $r \in |R| = \lim |R_i|$ be a point corresponding to an element of $\lim T_i$. Then r maps to (u, u') in $|U| \times |U|$ by construction and we see that $x = x'$ in $|X|$ as desired.

Parenthetical statement: A quasi-separated algebraic space is decent, see Decent Spaces, Section 03I7 (the key observation to this is Properties of Spaces, Lemma

03IJ). A locally separated algebraic space is decent by Decent Spaces, Lemma 088J. □